

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary End Semester Examination – Winter 2023

Course: B. Tech. Branch: Civil Engineering Semester: VI Semester

Subject Code & Name: BTCVOE605H Applications of Remote Sensing and GIS

Max Marks: 60

Date:25/01/2024

Duration: 3.00 Hr.

/Instructions to the Students:/

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

Q. 1 Solve Any Two of the following.

- A) Write short notes on: - **6**
- (a) Remote sensing platforms
- (b) Atmospheric windows
- (c) Electromagnetic radiation
- B) Write a detailed note on atmospheric scattering and its types. **6**
- C) Explain in detail about methods of remote sensing. **6**

Q.2 Solve Any Two of the following.

- A) Write short notes on: - **6**
- (a) Earth observation satellite
- (b) Swath
- (c) RADAR
- B) Write detailed note on spectral, spatial and temporal resolution. **6**
- C) Write a detailed note on microwave remote sensing technique and its applications. **6**

Q. 3 Solve Any Two of the following.

- A) Write short notes on: - **6**
- (a) Supervised classification of satellite imagery
- (b) Spectral signature of soil
- (c) Digital image
- B) Write a detailed note on the steps involved in digital image processing. **6**
- C) What are important keys in image interpretation and their importance? **6**

Q.4 Solve Any Two of the following.

- A) Write short notes on: - **6**
(a) Digital Elevation Model
(b) Cloud computing
(c) Triangular Irregular Network (TIN)
- B) Explain the working of Light Detection and Ranging (LiDAR) system with its applications. **6**
- C) Define and describe the Geographic Information System with its working. **6**

Q.5 Solve Any Two of the following.

- A) Write short notes on: - **6**
(a) Geoid
(b) Approximation of Earth
(c) Universal Transverse Mercator system
- B) Explain in detail about datum system and describe its types. **6**
- C) Explain Lambert Conformal Conic Projection system and its advantages and disadvantages. **6**

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